PREFACE

This book was based originally on a series of lectures, and is intended for readers who, while interested in natural sciences, have no previous training in theoretical physics and yet are familiar to a certain extent with physical ideas. In conformity with the express wish of the *Verband deutscher Elektrotechniker,*[a] under whose auspices the lectures were given, a short history of atomic physics, as well as a general review of contemporary knowledge of atomic and nuclear structure, are included here as an introduction. Obviously, a thorough understanding of nuclear physics cannot be gained from a short survey of this nature, but it may at least succeed in providing a basis for an understanding of the lectures on nuclear physics which follow. In my treatment of nuclear physics, I have departed somewhat from the method followed by other popular books on the subject, inasmuch as I have attempted to begin my discourse with the theory of the processes and reactions within the atom, and to discuss practical applications in conclusion only. At the same time, it was essential to make the theory intelligible without resort to mathematics, with the aid of illustrative models and by citing as analogies certain more widely known related phenomena. Nuclear physics lends itself to such a treatment more than many other branches of physical science. However, this method obviously has its natural limitations, and for a more profound understanding of the entire complex of relationships, a mathematical presentation of the subject is, of course, essential. For a thorough study of nuclear physics in this sense, there are many excellent books available. In the present volume, the technical apparatus of nuclear physics is discussed in the seventh chapter only; the eighth, and last, chapter presents a survey of the practical applications achieved up to the present time.

Since the publication, during the war, of the first edition of this book, reports have been published on the great progress in the field of nuclear physics, and especially on those technical developments relating to the atomic nucleus which had till then been restricted to the secret laboratories of the belligerent nations. These new developments are described, in general outline, in the last chapter of this book, where the practical applications of nuclear physics are discussed. Furthermore, those discoveries which were made or published after the war only, are dealt with in the text elsewhere.

The present English edition, appearing some time after the German one, may be of interest in connection with the history and the principles of nuclear physics rather than with respect to its recent development. Since the writing of the book and even since its last revision in 1948 an enormous development